The Magic of Exponentials: Growth, Decay, and Compounding

The Meenakshi School — Applied Mathematics & Science Series

Executive Overview

This lesson explores how the exponential function $y = a \cdot b^x$ governs fundamental phenomena across the Universe—from ancient algorithmic puzzles and subatomic physics to modern finance and cosmology.

1. Classical Growth: The Wheat and Chessboard Problem (Chaturanga)

- **The Premise:** The inventor of chess requested a seemingly modest reward from the King: 1 grain of rice on the first square, 2 on the second, 4 on the third, doubling on each subsequent square across the 64-square board.
- **The Mathematics:**

$$\text{Total Grains} = \sum_{k=0}^{63} 2^k = 2^{64} - 1 = 18,446,744,073,709,551,615$$

- **The Reality Check:** Over 18.4 quintillion grains of rice—equivalent to nearly **3,000 years of global agricultural production**.

2. Exponential Decay: Medical Physics & Nuclear Tracers

- **The Concept:** While growth multiplies by $b > 1$, decay scales down by $0 < b < 1$. Nuclear medicine uses short-lived radioactive isotopes like **Technetium-99m ($^{99\mathrm{m}}\mathrm{Tc}$)** for SPECT imaging because they clear quickly from the human body.
- **The Decay Equation:**

$$N(t) = N_0 \left(\frac{1}{2}\right)^{\frac{t}{t_{1/2}}}$$

  - N_0: Initial administered dose
  - $t_{1/2}$: Half-life ($6 \text{ hours for } ^{99\mathrm{m}}\mathrm{Tc}$)
  - t: Elapsed time in hours
- **Class Exercise:**
  - *Question:* If a patient receives a dose at 8:00 AM, how much remains in the body at 8:00 AM the following day (24 hours later)?
  - *Calculation:* $t/t_{1/2} = 24/6 = 4 \text{ half-lives.}$
  - *Result:* $\left(\frac{1}{2}\right)^4 = \frac{1}{16} = 6.25\%$ remaining.

3. Financial Compounding: The 10 Paise Doubling Effect

- **The Scenario:** Compare making a single large deposit versus doubling a tiny initial amount of $10 \text{ paise } (0.10 \text{ ₹})$ every day for 30 days.
- **Day 30 Single-Day Value:**

$$\text{Value}_{\text{Day 30}} = 0.10 \times 2^{29} = 0.10 \times 536,870,912 = 53,687,091.20 \text{ ₹}$$

- **Cumulative 30-Day Account Balance:**

$$\text{Total Balance} = \sum_{k=0}^{29} (0.10 \times 2^k) = 0.10 \times (2^{30} - 1) = 107,374,182.30 \text{ ₹}$$

- **Takeaway:** Over 107 Million ₹ (or 10.74 Crores) accumulated from a starting balance of just 10 paise!

4. Universal Comparative Summary

Application Domain	Mathematical Model	Base Behavior (b)	Real-World Impact
Ancient Algorithmic Growth	$y = 2^x$	$b = 2 > 1$	Outstrips global planetary supply (18.4 quintillion)
Nuclear Medicine (Decay)	$y = \left(\frac{1}{2}\right)^{x/6}$	$b = 0.5 < 1$	Clears 93.75% of tracer radiation within 24 hours
Financial Compounding	$y = 0.10 \cdot (2)^x$	$b = 2 > 1$	Converts 10 paise into $>$ 10.7 Crores in 30 days
Cosmological Redshift	$\lambda_{\text{obs}} = \lambda_{\text{rest}}(1 + z)$	$1 + z = 1.158$	Shifts photons 2.4B light-years across expanding space